Salva Ardid, PhD.

Research Positions

2020-07 — **PI Group Leader**
Universitat Politècnica de València — València, Spain

2020-06 | 2019 — **Associate Research Scientist**
Yale University — New Haven CT, USA

- Advisor: I Levy

2019 | 2018 — **Visiting Research Scientist**
Boston University — Boston MA, USA

- Advisor: N Kopell

2018 | 2008 — **Postdoctoral Research Scientist**
Boston University | York University | Yale University — USA & Canada

- Advisor: N Kopell (BU) | T Womelsdorf (YorkU) | XJ Wang (Yale)
- Period: 2018-2014 2014-2011 2011-2008

2008 | 2003 — **PhD Candidate Scientist**
Neuroscience Institute UMH-CSIC — Alacant, Spain

- Advisor: A Compte

Education

2008 — **PhD in Computational Neuroscience with Honors**
Neuroscience Institute, UMH-CSIC — Alacant, Spain

- Thesis: Neural Mechanisms of Selective Attention in a Microcircuit Model of Interacting Cortical Areas
- PhD Advisor: A Compte

2003 — **MEng in Electronic Engineering with Honors**
Universitat de València — València, Spain

1999 — **BSc + MSc in Physics**
Universitat de València — València, Spain

Main Publications

2019 — **Feature-specific prediction errors and surprise across macaque fronto-striatal circuits**
Nature communications

- M Oemisch, S Westendorff, M Azimi, SA Hassani, S Ardid, P Tiesinga, ...
- Citations: 9

Computational Neuroscientist

Dept. of Comparative Medicine
Yale University

salva.ardid@yale.edu
Scholar
GitHub
Bitbucket
Twitter

Computing Proficiency

- Bash
- C++
- GNU R
- LaTeX
- Linux/HPC
- MATLAB/GNU Octave
- Python
- Version Control

Open Source Contributions

`dynaSim`
Toolbox for modeling dynamical systems (MATLAB/Octave)

`fig2svg`
Code to export Matlab/Octave figures to SVG format

`SierraBreezeEnhanced`
Enhanced KDE Breeze Window Decorations

`waveformAnalysis`
Code to identify broad vs. narrow neurons based on action potential waveforms (MATLAB/Octave)

`clusteringAnalysis`
Code for functional cell classification based on waveform and firing properties (MATLAB/Octave)

Languages

Catalan: native
English: fluent
Spanish: native

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Last updated on 2020-02-08.

2019 **Biased competition in the absence of input bias revealed through corticostriatal computation**
Proceedings of the National Academy of Sciences

- S Ardid, JS Sherfey, MM McCarthy, J Hass, BR Pittman-Polletta, N Kopell
- Citations: 2

2018 **DynaSim: a MATLAB Toolbox for neural modeling and simulation**
Frontiers in neuroinformatics

- JS Sherfey, AE Soplata, S Ardid, EA Roberts, DA Stanley, ...
- Citations: 13

2018 **Flexible resonance in prefrontal networks with strong feedback inhibition**
PLoS computational biology

- JS Sherfey, S Ardid, J Hass, ME Hasselmo, NJ Kopell
- Citations: 8

2017 **A computational psychiatry approach identifies how alpha-2A noradrenergic agonist Guanfacine affects feature-based reinforcement learning in the macaque**
Scientific reports

- SA Hassani, M Oemisch, M Balcarras, S Westendorff, S Ardid, ...
- Citations: 11

2016 **Attentional selection can be predicted by reinforcement learning of task-relevant stimulus features weighted by value-independent stickiness**
Journal of cognitive neuroscience

- M Balcarras, S Ardid, D Kaping, S Everling, T Womelsdorf
- Citations: 10

2016 **Neuroswarm: a methodology to explore the constraints that function imposes on simulation parameters in large-scale networks of biological neurons**
Uncertainty in biology

- D Gomez-Cabrero, S Ardid, M Cano-Colino, J Tegnér, A Compte
- Citations: 5

2015 **Anterior cingulate cortex cells identify process-specific errors of attentional control prior to transient prefrontal-cingulate inhibition**
Cerebral Cortex

- C Shen, S Ardid, D Kaping, S Westendorff, S Everling, T Womelsdorf
- Citations: 36

2015 **Mapping of functionally characterized cell classes onto canonical circuit operations in primate prefrontal cortex**
Journal of Neuroscience

- S Ardid, M Vinck, D Kaping, S Marquez, S Everling, T Womelsdorf
- Citations: 35

2014 **Burst firing synchronizes prefrontal and anterior cingulate cortex during attentional control**
Current Biology

- T Womelsdorf, S Ardid, S Everling, TA Valiante
- Citations: 63

2014 **"Adaptive learning" as a mechanistic candidate for reaching optimal task-set representations flexibly**
BMC neuroscience

- S Ardid, M Balcarras, T Womelsdorf
- Citations: 1

2013 **A tweaking principle for executive control: neuronal circuit mechanism for rule-based task switching and conflict resolution**
Journal of Neuroscience

- S Ardid, XJ Wang
- Citations: 27

2013 **Subnetwork selection in deep cortical layers is mediated by beta-oscillation dependent firing**
Frontiers in systems neuroscience

- T Womelsdorf, S Westendorff, S Ardid
- Citations: 2

2011 **What can tracking fluctuations in dozens of sensory neurons tell about selective attention?**
Frontiers in systems neuroscience

- JD Murray, S Ardid
- Citations: 2

2010 **Reconciling coherent oscillation with modulation of irregular spiking activity in selective attention: gamma-range synchronization between sensory and executive cortical areas**
The Journal of Neuroscience

- S Ardid, XJ Wang, D Gomez-Cabrero, A Compte
- Citations: 53

2007 **An integrated microcircuit model of attentional processing in the neocortex**
Journal of Neuroscience

- S Ardid, XJ Wang, A Compte
- Citations: 100

Scholar Profile

378 Citations
As of 2020-02-08

- h-index: 9
- i10-index: 9

Main Talks & Seminars

2020 **Biological Sciences Training Program in Psychiatry**
Yale University — New Haven CT, USA

2019 **Math Biology Seminar Series**
Brandeis University — Waltham MA, USA

2019 **School of Psychology**
University of Birmingham — Birmingham, UK

2018 **Southwestern Medical Center**
University of Texas
Dallas TX, USA

2017 **Department of Psychology**
Vanderbilt University
Nashville TN, USA

2017 **Icahn School of Medicine**
Mount Sinai
New York City NY, USA

2017 **Department of Neurobiology & Behavior**
Stony Brook University
Stony Brook NY, USA

2016 **5th BARCCSYN Meeting**
Barcelona Computational, Cognitive, and Systems Neuroscience Community
Barcelona, Spain

2016 **Rhythmic Dynamics and Cognition Symposyum**
Cognitive Rhythms Collaborative
Boston MA, USA

2011 **IDIBAPS**
Hospital Clinic
Barcelona, Spain

2010 **Robarts Research Institute**
Western University
London ON, Canada

2009 **Kavli Workshop on Decision Making**
Yale University
New Haven CT, USA

2007 **12th SENC Congress**
Sociedad Española de Neurociencia
València, Spain

2007 **Seminars in Biophysics/Biomathematics**
Universitat Politècnica de Catalunya
Barcelona, Spain

2006 **15th Annual Computational Neuroscience Meeting**
CNS* Featured Talk
Edinburgh, UK

Teaching, Mentoring & University Services

2019 | 2018 **Master Advisor**
Deep Learning Applied to Particle Physics, M Ortiz
València, Spain

- Master in Artificial Intelligence
- Universidad Internacional de Valencia (VIU)

2019 | 2018 **Member of the Master Thesis Evaluation Committee**
Number of Thesis Defenses: 6
València, Spain

- Master in Artificial Intelligence
- Universidad Internacional de Valencia (VIU)

2015 **Linear Algebra**
Boston University
Boston MA, USA

2014 **Guest Lecture: Dynamical Systems Models of Selective Attention**
Massachusetts Institute of Technology — Boston MA, USA

- Course: Understanding Visual Attention through Computation, J Tsotsos
- Department of Brain and Cognitive Sciences

2003 | 2001 **High School Teacher**
Department of Education — Catalonia, Spain

Organization of Events

2020 **Computational Psychiatry and Neuroeconomics: Why and How?**
Seminar Series — New Haven CT, USA

- Organizers: S Ardid, A Powers, H Pushkarskaya
- Yale University

2012 **Understanding Heterogeneous Cortical Activity: The Quest for Structure and Randomness**
COSYNE Workshop — Snowbird Resort UT, USA

- S Ardid, A Bernacchia, T Engel

Awards, Grants & Fellowships

2024 | 2020 **Talented Researcher Support Programme**
GenT Plan Award — València, Spain

- Principal Investigator
- Amount: 381,500 EUR

2018 | 2017 **Army Research Office Grant**
Boston University — Boston MA, USA

- co-PI
- Amount: 120,000 USD

2011 | 2008 **Swartz Fellowship in Computational and Systems Neuroscience**
Yale University — New Haven CT, USA

2008 | 2004 **Volkswagen Foundation Fellowship**
Neuroscience Institute UMH-CSIC — Alacant, Spain

2007 **Barcelona Supercomputing Center Fellowship**
Barcelona Supercomputing Center — Barcelona, Spain

2004 | 2003 **Universidad Miguel Hernandez Fellowship**
Neuroscience Institute UMH-CSIC — Alacant, Spain